

BACHELOR WITH COMPUTER APPLICATIONS AS MAJOR (CT-1)
6th SEMESTER

CAP622J1: COMPUTER APPLICATIONS _ PYTHON PROGRAMMING

CREDITS: THEORY (3) PRACTICAL (1)

LEARNING OUTCOMES:

After completing this course, the learner shall be able to:

1. *Understand the fundamentals of programming through Python*
2. *Transform a solution from a subjective world into an objective world.*
3. *Work with and manipulate different basic data structures available in Python*
4. *Use the List comprehensions and generators in their programs*
5. *Apply basic object-oriented concepts to design classes and objects in Python*
6. *Use concepts of Inheritance and Polymorphism in their programs*
7. *Perform basic file operations for text and CSV files*
8. *Use existing inbuilt Python modules*
9. *Develop custom modules*
10. *Use basic functionalities provided by packages Numpy and Pandas*
11. *Develop visualizations using different plotting functions available in matplotlib*

UNIT-I: (15 LECTURES)

Origins and History of Python, Structure of a Python Program, Interpreter shell, Indentation, Comments, Identifiers and keywords, Literals, Basic operators (Arithmetic operator, Relational operator, Logical or Boolean operator, Assignment Operator, Bit wise operator)

Entering Expressions into the Interactive Shell, The Integer, Floating-Point, and String Data Types, String Concatenation and Replication, Storing Values in Variables. Creating Python Programs: Input and Output Statements, Control statements: -branching, looping, range function, Exit function, break, continue and pass.

UNIT: II (15 LECTURES)

Functions: Defining a Function, Calling a Function, def Statements with Parameters, Formal and Actual Arguments, Positional Arguments, Keyword Arguments, Default Arguments, Variable Length Arguments, Return Values and return Statements, The None Value, Local and Global Scope, The global Statement.

The List Data Type: Creating Lists, Basic List Operations, Indexing and Slicing in Lists, Built-In Functions Used on Lists, List Methods. List comprehension.

UNIT: III (15 LECTURES)

The Dictionary Data Type: Creating Dictionary, Accessing and Modifying key: value Pairs in Dictionaries, Built-In Functions used on Dictionaries, Dictionary Methods

Tuples and Sets, Creating Tuples, Basic Tuple Operations, Indexing and Slicing in Tuples, Built-In Functions Used on Tuples, Set Methods, Traversing of Sets

Strings, Creating and Storing Strings, Basic String Operations, Accessing Characters in String by Index Number, String Slicing and Joining, String Methods, Formatting Strings.

Files: Types of Files, Creating and Reading Text Data

TEXT BOOKS

1. Gowrishankar S, Veena A, "Introduction to Python Programming", 1st Edition, CRC Press/Taylor & Francis, 2018. ISBN-13: 978-0815394372
2. Downey, A.B., (2015), Think Python How to think like a Computer Scientist, 3rd edition.
3. Taneja, S. & Kumar, N., (2017), Python Programming- A Modular Approach. Pearson Education.

REFERENCE BOOKS

1. Jake VanderPlas, "Python Data Science Handbook: Essential Tools for Working with Data", 1st Edition, O'Reilly Media, 2016. ISBN-13: 978-1491912058
2. Aurelien Geron, "Hands-On Machine Learning with Scikit-Learn and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems", 2nd Edition, O'Reilly Media, 2019. ISBN – 13: 978-9352139057.
3. Wesley J Chun, "Core Python Applications Programming", 3rd Edition, Pearson Education India, 2015. ISBN-13: 978-9332555365

4. Miguel Grinberg, “Flask Web Development: Developing Web Applications with Python”, 2nd Edition, O'Reilly Media, 2018. ISBN-13: 978-1491991732.
5. Brown, M. C. (2001). The Complete Reference: Python, McGraw Hill Education.
6. Dromey, R. G. (2006), How to Solve it by Computer, Pearson Education.
7. Gutttag, J. V. (2016), Introduction to computation and programming using Python. MIT Press.
8. Liang, Y.D. (2013), Introduction to programming using Python. Pearson Education.

LAB WORK: PYTHON PROGRAMMING

1. Install and configure Python IDE
2. Write simple Python program to display message on screen
3. Write a program to demonstrate different number data types & arithmetic operations in Python.
4. Write a Python program to demonstrate use of conditional statements:
5. Write Python program to demonstrate use of looping statements
6. Demonstrate the following control transfer statements in Python with suitable examples.
 - i) break
 - ii) continue
 - iii) pass
7. Write a python program to demonstrate the working of following functions in Python.
 - i) id()
 - ii) type()
 - iii) range()
8. Write Python programs to demonstrate the following:
 - i) input()
 - ii) print()
 - iii) 'sep' attribute
 - iv) 'end' attribute
9. Write a program to create a menu driven program with the following options using functions:
Addition 2. Subtraction 3. Multiplication 4. Division
10. Write Python program to perform following operations on Lists:
 - a) Create list
 - b) Access list
 - c) Update list (Add item, Remove item)
 - d) Delete list
11. Write Python program to perform following operations on Tuples:
 - a) Create Tuple
 - b) Access Tuple
 - c) Update Tuple
 - d) Delete Tuple
12. Write Python program to perform following operations on Sets:
 - a) Create Set
 - b) Access Set elements
 - c) Update Set
 - d) Delete Set
13. Write Python program to perform following operations on Dictionaries:
 - a) Create Dictionary
 - b) Access Dictionary elements
 - c) Update Dictionary
 - d) Delete Set
 - e) Looping through Dictionary
14. Write a Python program to demonstrate slicing operations on lists and strings.
15. Write a Python program to perform read and write operations on a file.